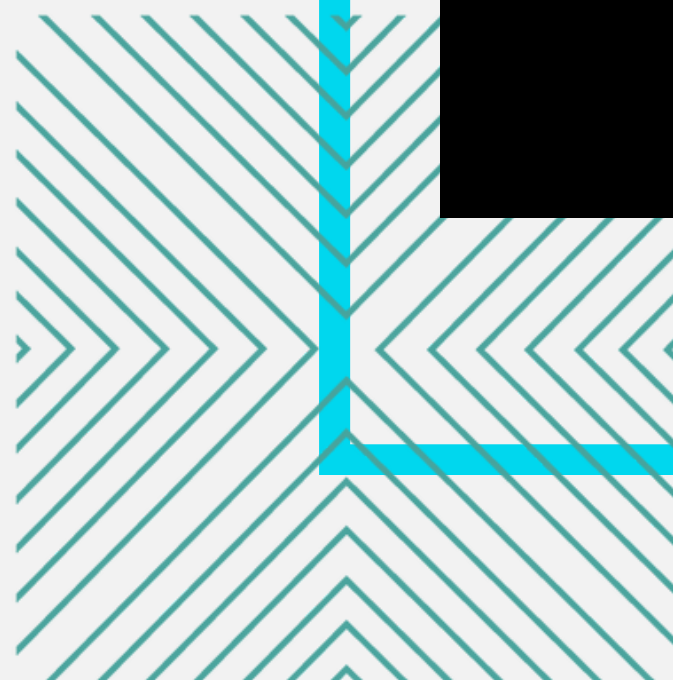
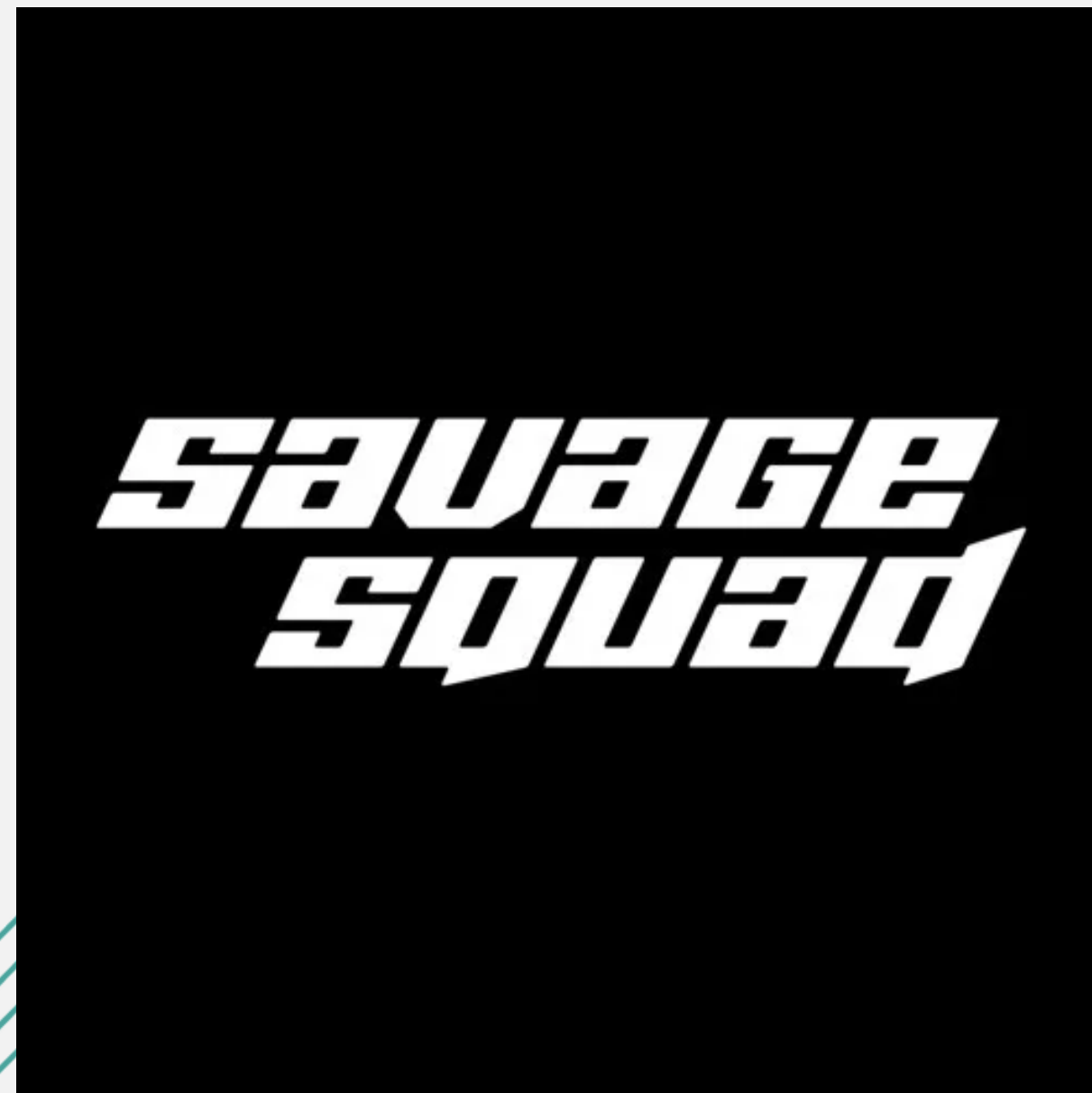




2303-Capstone-Group-A

Savage Squad





Introduction

We are Group A of Capstone Project Cohort 2 named as "Savage Squad". Team of 5 Data Engineers with each and every member fully enthusiastic about learning and implementing what is taught. Introducing our Team members:



Mehboob Ali
(Team Lead)

Ali Umair

Muhammad Arsalan

Rahima Siddiqui

Talha Khan



Problem Statement

Accessing the right healthcare provider with specialized expertise remains a significant challenge in the current healthcare landscape. Patients often struggle to find practitioners who can effectively address their specific health concerns, leading to sub optimal treatment outcomes and frustrating experiences. To address this issue, there is a need for an advanced patient-provider matching system that leverages patient-provided information, provider success metrics, and local availability to suggest suitable practitioners to patients.

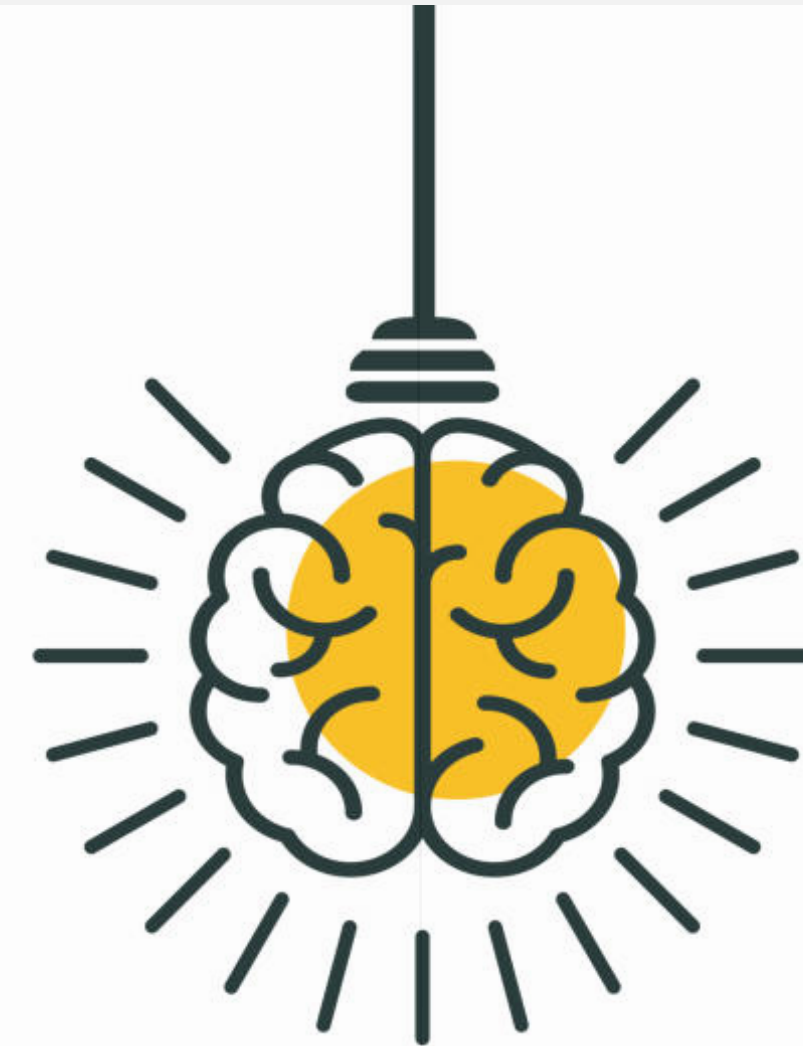




Patient to Provider Matching Service

Proposed Solution

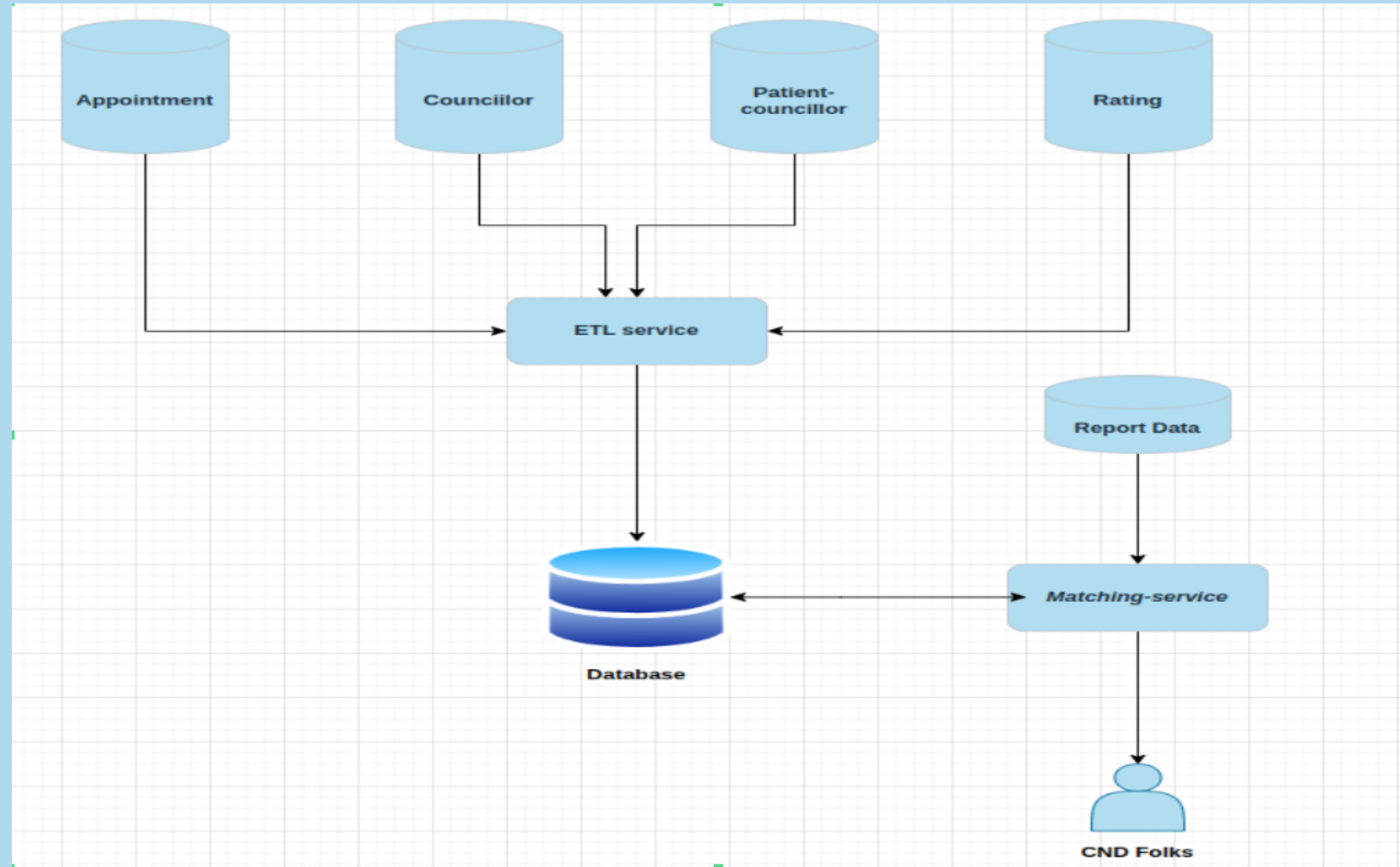
Our proposed solution aims to develop an intelligent patient-provider matching service that revolutionizes the way patients find suitable healthcare professionals. By combining patient-provided information and provider success metrics the system will offer personalized recommendations adjusting to each patient's unique needs. The system will consider factors such as specialization, treatment success rates, patient reviews to facilitate informed decision-making and streamline the healthcare access process.



Approach



Architectural Diagram



01

Data coming from four APIs.

02

ETL Service.

03

Loading Data to Redis.

04

Matching Service.

FastApi web app that will give councilors based on patient report data and redis database.



Methodology



Tech Stack



Python



Redis



Apache
Spark



FastApi



Docker



GitHub

ETL Service



Extract

Extraction of Data using four endpoints.

- Appointment
- Councilor
- Patient-Councilor
- Rating



Transform

Transformation of Data that is given by extract phase. All the four APIs were joined in transformation phase to represent the needed data.

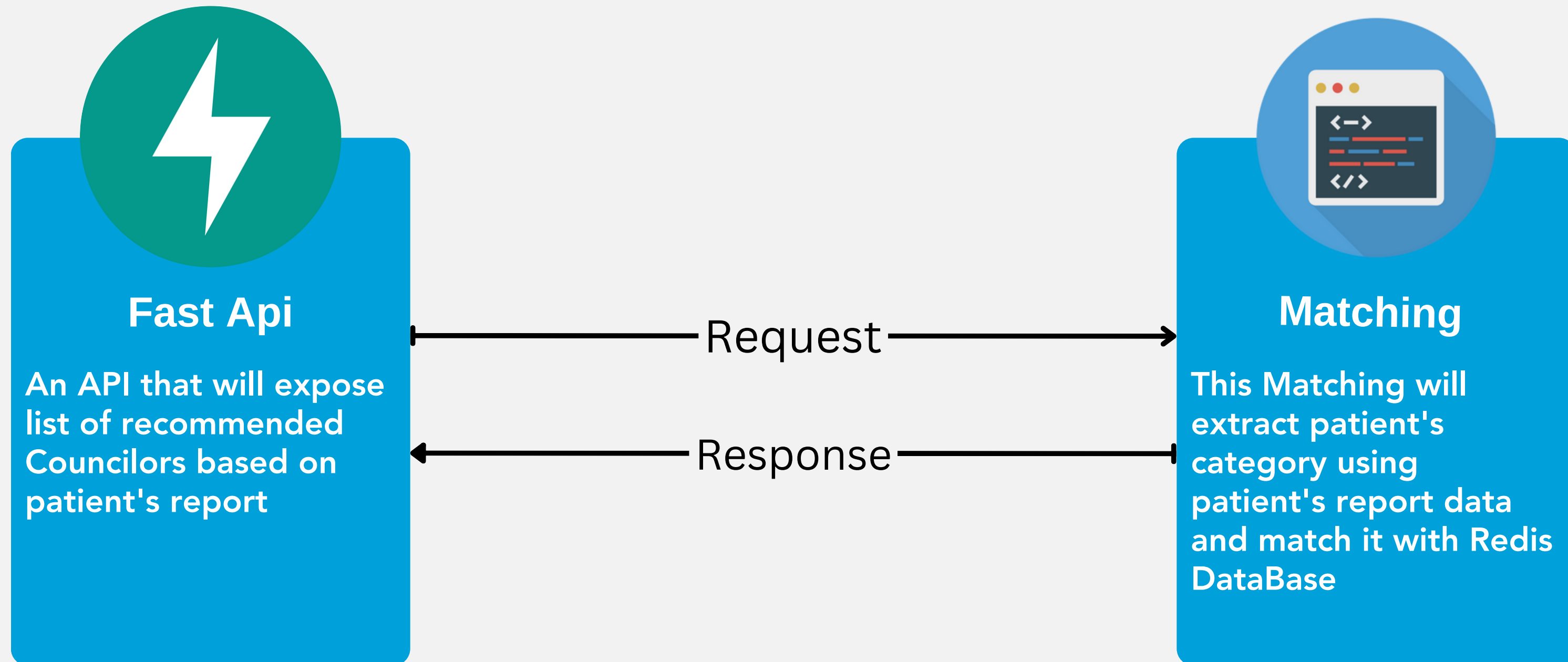


Load

Once the Data is transformed it is loaded to Redis in the load phase.



Matching Service



Challenges

Technical

01. Learning Curve for new technologies.
02. Scheduling.
03. Following Best Practices.
04. Version Control.
05. Different approaches.

Non-Technical

01. Time Management.
02. Team Communication.
03. Decision Making.
04. Balancing team goals.
05. Managing Expectations.



Q&A



Thank you for your Attention!

